RESEARCH METHODS

DAY 3

Welcome Back to Day 3!

Today we shape your research projects.

First, lets talk about your homework.

Handout / Homework

What we know/assume
What we want to know

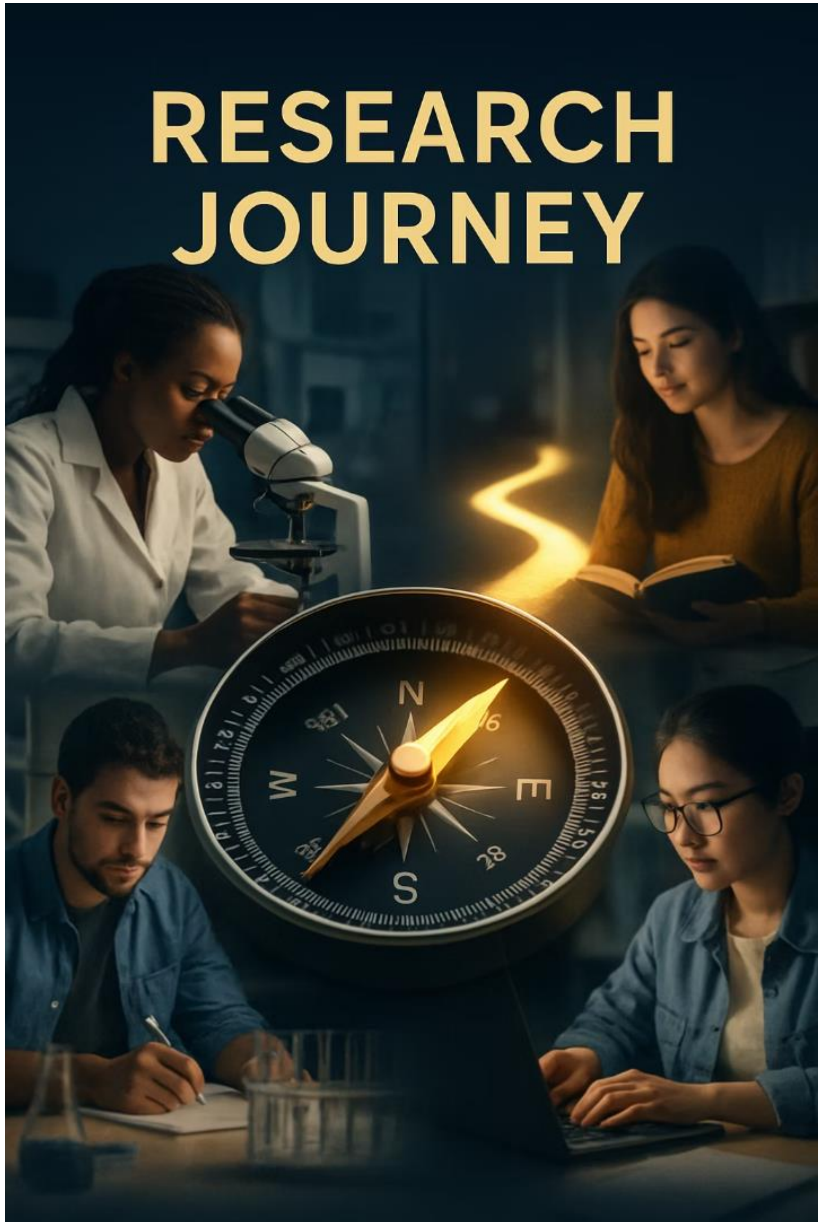
Narrowed Down Research Questions &
BRIC-FOR Ratings

Good reminder:

> *"The formulation of a problem is often more essential than its solution, which may be merely a matter of mathematical or experimental skill."*

Albert Einstein

The quality of your question largely determines the quality of your research. Take time to craft brick-strong (BRIC-FOR) questions.



The Research Process

The journey from question to insight follows these essential steps:

Ask a Question
Formulate a clear, specific question worth investigating

Gather Information
Review the literature & collect relevant data through appropriate methods

Analyze the Information
Examine data for patterns, trends, and meaning

Draw Conclusions
Develop evidence-based answers to your original question

Share Findings
Communicate results effectively to appropriate audiences

NOW LET'S TURN OUR ATTENTION TO THE NEXT P

Gathering Information through

LITERATURE REVIEW

Efficient Literature Search

Smart Keywords

- Choosing keywords that would help you find the right content.
- Try synonyms.
- Use quotation marks "Climate Change"

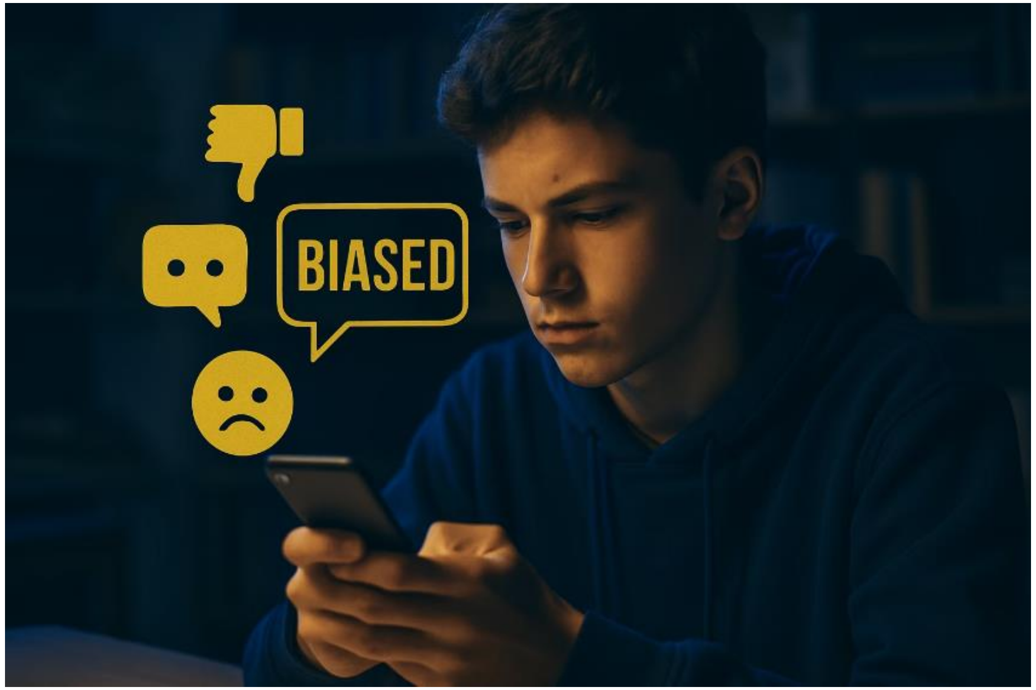
Intentional Sources

- Google Scholar - if available
- Library access to databases (and use filters)
- Reliable sites with .edu .gov or trusted organizations

Initial Reading

- Abstract - summary to read
- Conclusions section
- Review the graphs
- Track and maybe "rate" what you find using a "Source Tracker".

How to Spot Credible Resources



- Check the Author - Is the author an expert in the field? Look for credentials or name of the university or organization.

- Look at the Source - Prefer .edu, .gov, and well-known .org or .com sites (like National Geographic, Mayo Clinic). Avoid personal blogs or random forums.

- Check the Date - Is the info recent and up-to-date? For most topics, aim for the last 5–10 years.

- Is It Cited? Does it give references?
Good sources list where their info comes from (references, links, or footnotes).

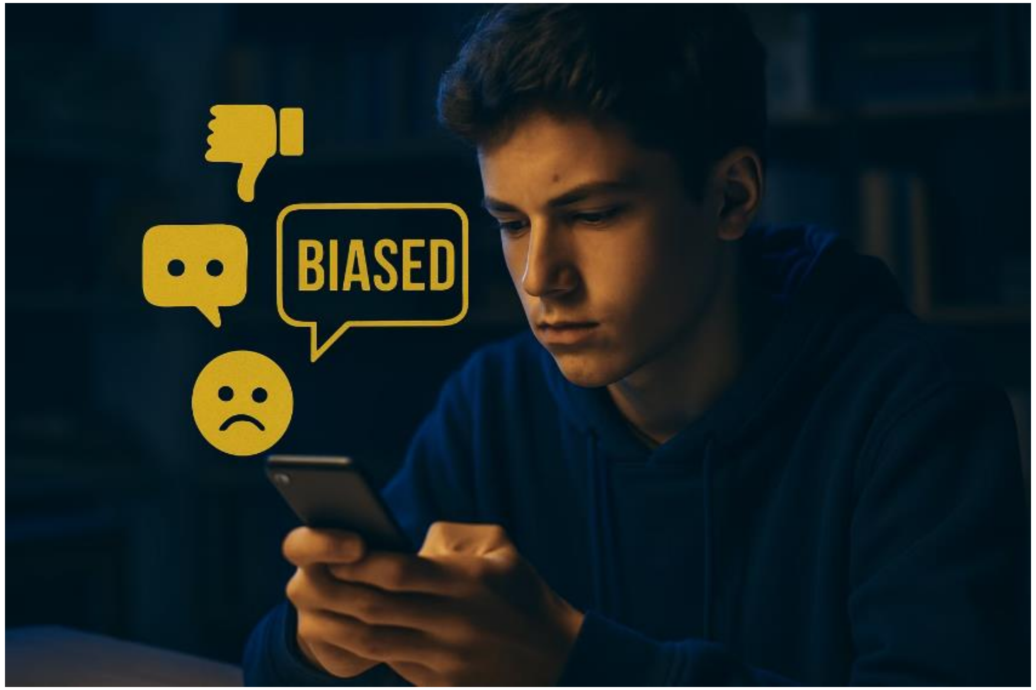
How to Spot Credible Resources

- Bias Alert - Does the source present facts or just opinions? Watch for emotional language or one-sided arguments.

- Peer-Reviewed? - For academic research, articles from journals that have been reviewed by other experts are best (Google Scholar is great for this).

- Design + Grammar - Trustworthy sites look professional and are free of spelling mistakes or sketchy ads.

- Cross-Check -
See if the same info appears in multiple trusted sources. That's a good sign it's reliable.



Activity: Literature Hunting!

Let's see what other researchers have found about your issue?

Use your **Source Tracker** doc!

NEXELLENCE INSTITUTE

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Initial Reading

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More articles to come...

Soon, we will share a package of articles that you can read which would facilitate your preparation process.

But feel free to explore more on your own!

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From Questions to Hypotheses

The information you have collected from the literature may help you translate your questions to hypotheses.

Singular: Hypothesis
Plural: Hypotheses

From Questions to Hypotheses

What is a Hypothesis?

An educated guess about what you expect to find based on your research question.

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What is a Hypothesis?

An educated guess about what you expect to find based on your research question.

A hypothesis transforms your curiosity into a testable prediction that guides your research process.

Good hypotheses are developed from prior knowledge, preliminary research, and logical reasoning.

Characteristics of a Good Hypothesis

It's a statement, not a question

Directly answers your research question or predicts an outcome.

It is testable

You can gather data to support or refute it.

It predicts a relationship between variables

Often phrased as "If X happens, then Y will happen."

Example: Study Time Hypothesis

Research Question

"How does the amount of time spent studying affect test scores among 8th graders?"

Hypothesis

"8th graders who study for at least 2 hours a day will have higher test scores than those who study for less than 1 hour a day."

It's a statement

It's a statement predicting an outcome (more study → higher scores).

It's testable

You could gather data on study habits and test scores.

It mentions specific groups to compare

Those who study ≥2 hours vs. those who study <1 hour.

Example: Sleep and Sugary Drinks

Research Question

"Does drinking sugary drinks before bed affect sleep quality in teenagers?"

Hypothesis

"Drinking sugary drinks in the evening will lead to poorer sleep quality in teenagers, with more frequent waking during the night."

- **It's a statement**
- **It's testable** through comparison of sleep patterns
- **It mentions a specific group**
- It **predicts** an effect (poorer sleep) and gives a **rationale** (sugar disrupts sleep)

Variables in Research

Independent Variable

What you change or what differs between groups.

The potential cause or input you're testing.

Example: amount of study time

Dependent Variable

What you measure as the outcome.

The effect that might change based on the independent variable.

Example: test scores

Hypothesis Formula

"If [independent variable], then [dependent variable], because [reason]."

Example: Study Time Hypothesis

Research Question

"How does the amount of time spent studying affect test scores among 8th graders?"

Hypothesis

"8th graders who study for at least 2 hours a day will have higher test scores than those who study for less than 1 hour a day."

Independent variable: Study Time
Dependent variable: Test scores

Example: Sleep and Sugary Drinks

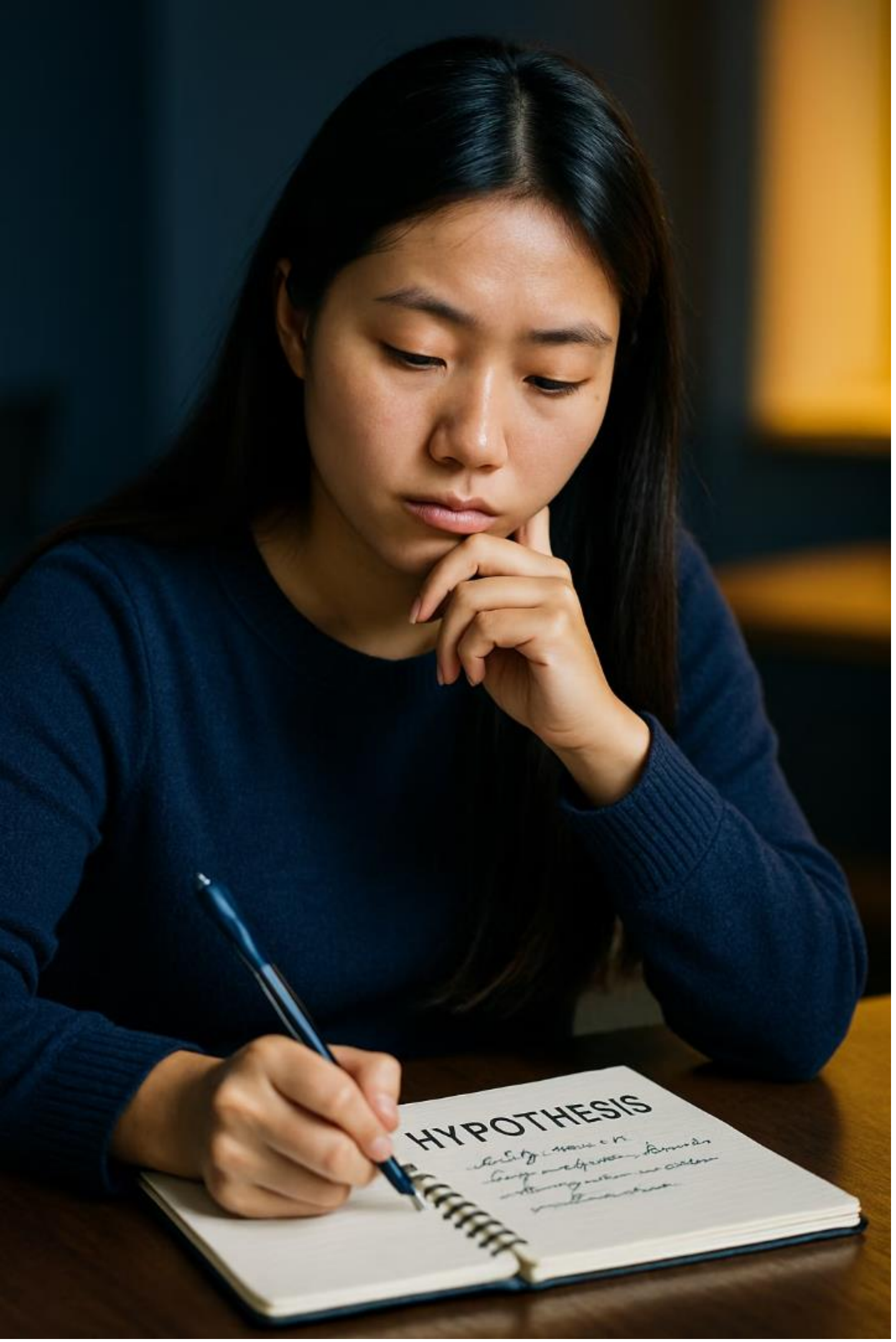
Research Question

"Does drinking sugary drinks before bed affect sleep quality in teenagers?"

Hypothesis

"Drinking sugary drinks in the evening will lead to poorer sleep quality in teenagers, with more frequent waking during the night."

Independent variable: Drinking sugary drinks
Dependent variable: Sleep quality, more specifically frequency of night awakenings



Your Turn: Formulate a Hypothesis

1. **Make a Prediction**

   Based on what you know, what do you expect to find?

2. **Write as a Statement**

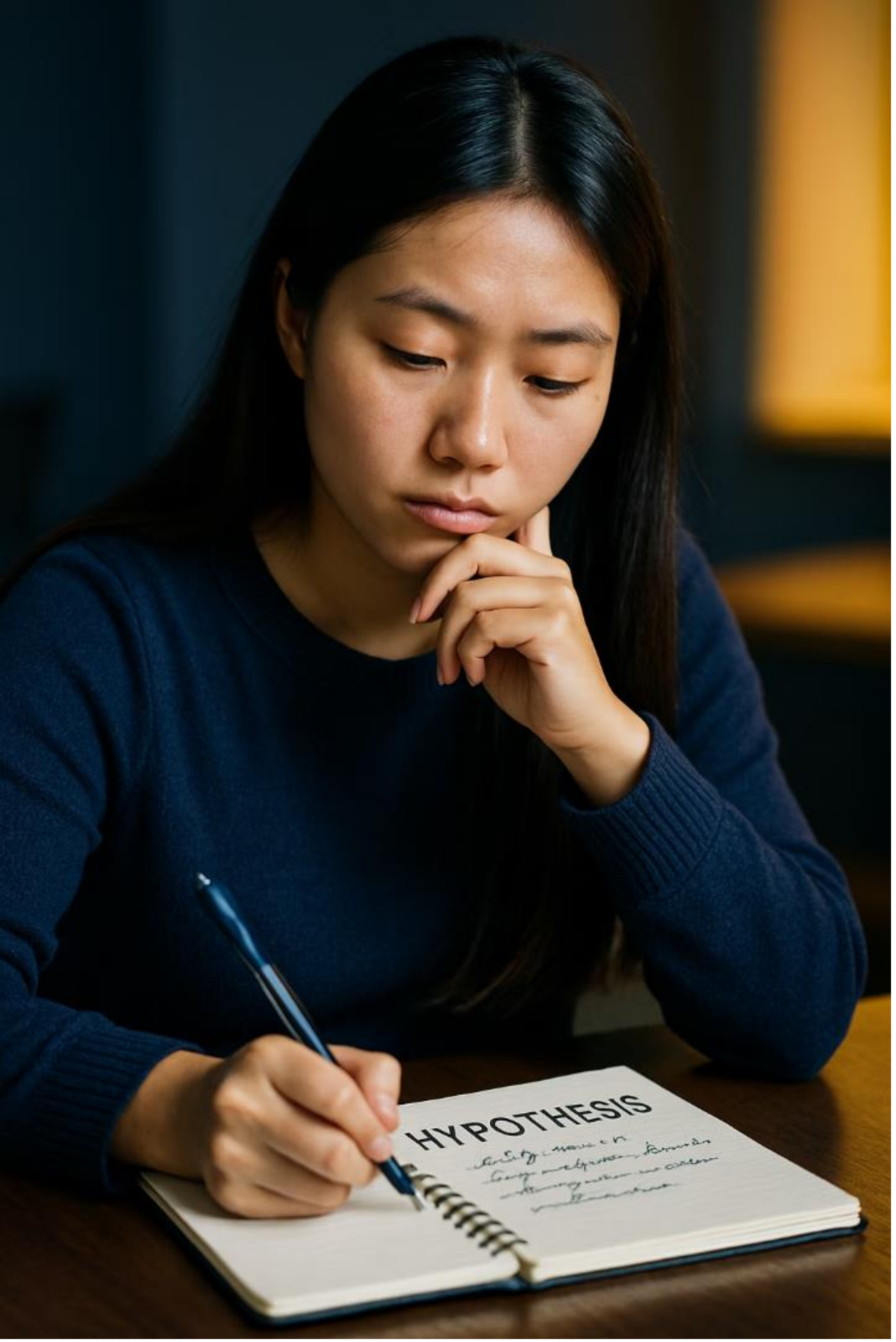
   Not a question - a clear prediction of what will happen.

3. **Identify & Include Variables**

   Connect your independent and dependent variables.

4. **Add Reasoning**

   Explain why you expect this outcome ("because...").



Your Turn:
Share Your Hypothesis

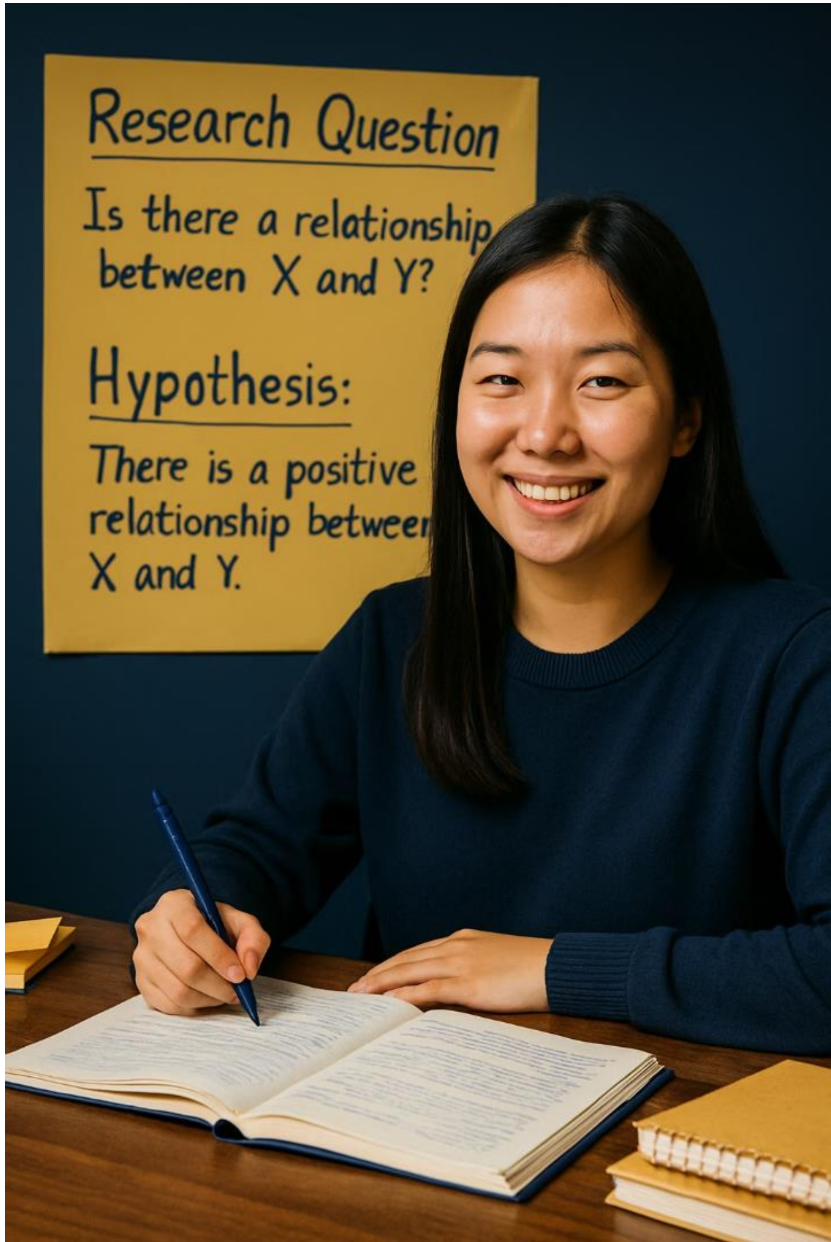
- Is it relevant to the issue at hand?
- Is it a statement?
- Is it testable?
- Does it predict a relationship between variables?
- Is it clear?
- Does it mention who?

REFINEMENT IS NORMAL!

Questions and hypotheses often evolve as you learn more.

But how do you “learn more”?

Answer: Literature Review



Key Takeaways

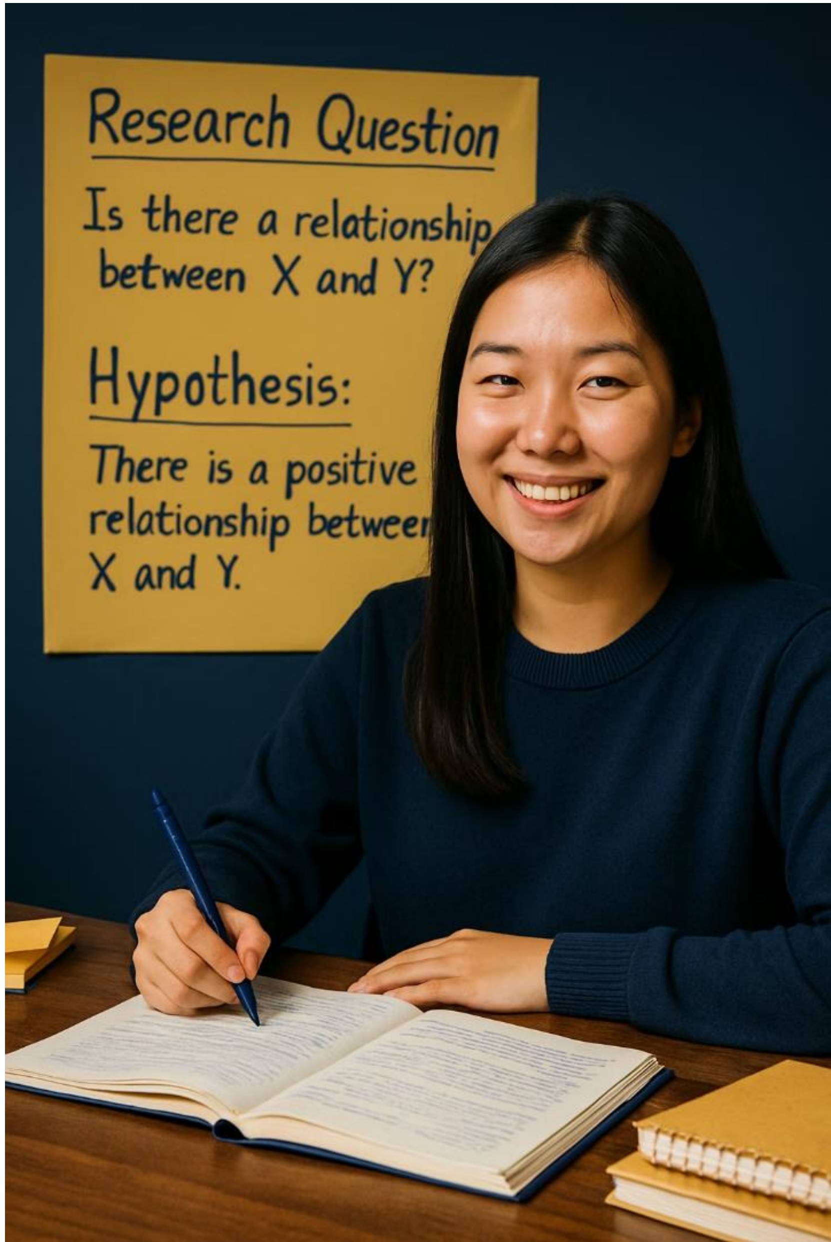
Good Questions Drive Good Research

Narrowing Is Essential

Hypotheses Are Educated Guesses

Refinement Is Normal

Literature is a source of information.
How you approach it is critical!



NEXT

Designing a Research Project

Designing a Research Project

Let's explore how to plan, execute, and analyze research that answers meaningful questions.

Failing to Plan is Planning to Fail

Research Requires Direction

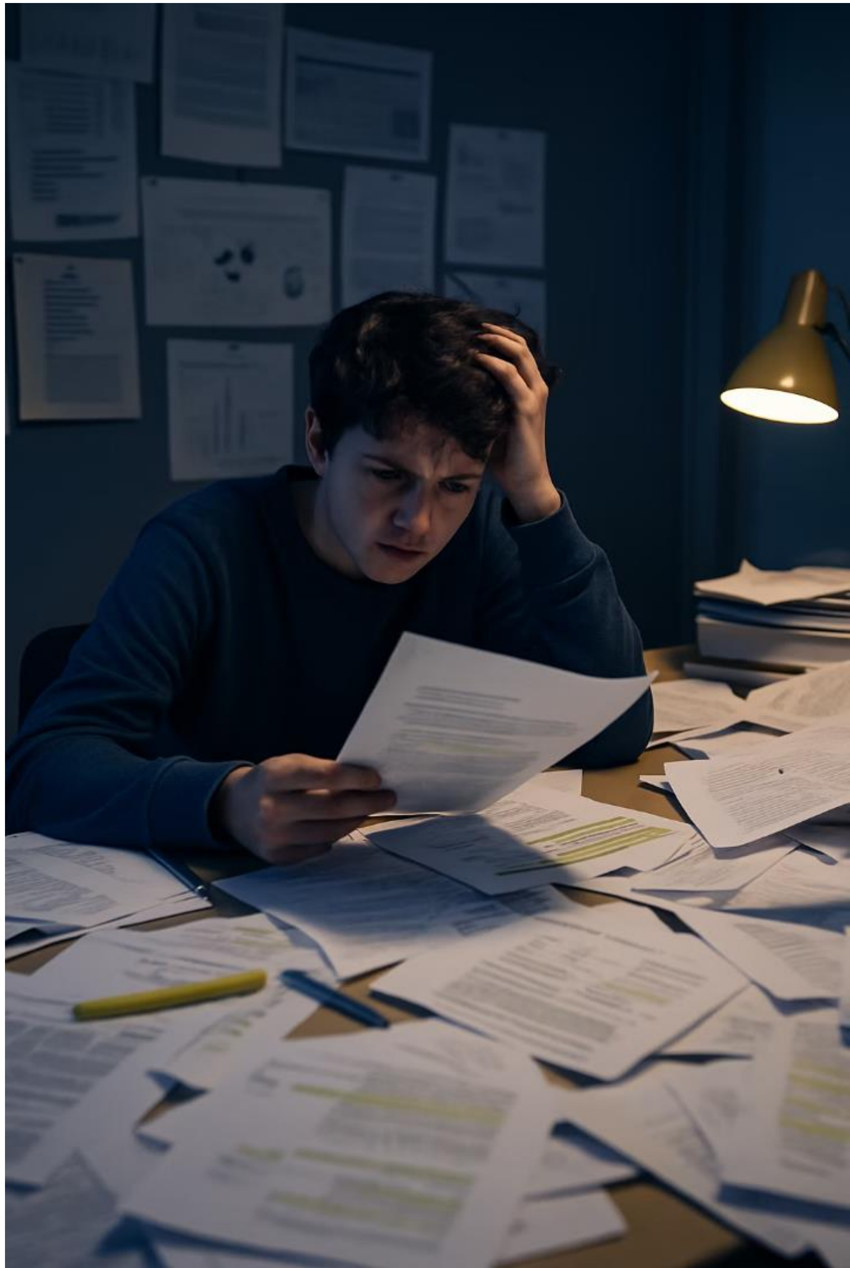
Like baking without a recipe or traveling without a map, research without planning leads to confusion.

Create a Blueprint

A well-designed project saves time and increases the chances of valid, meaningful findings.

Develop a Game Plan

We'll cover methodologies, tools, variables, and ethical considerations for your research journey.



Aisha's Research Journey

Great Question

Aisha wondered if music affects homework productivity.

Poor Planning

She jumped into gathering data without specifying music type or how to measure productivity.

Back to Drawing Board

Confused by inconsistent results, Aisha realized she needed a better plan.

Improved Design

She created a controlled study with specific tasks, consistent music, and measurable outcomes.

Choosing a Research Method

The first step in designing your project is selecting the right method to answer your question.

Methodology includes…

- Your overall approach / strategy:

- Your data collection tools / techniques

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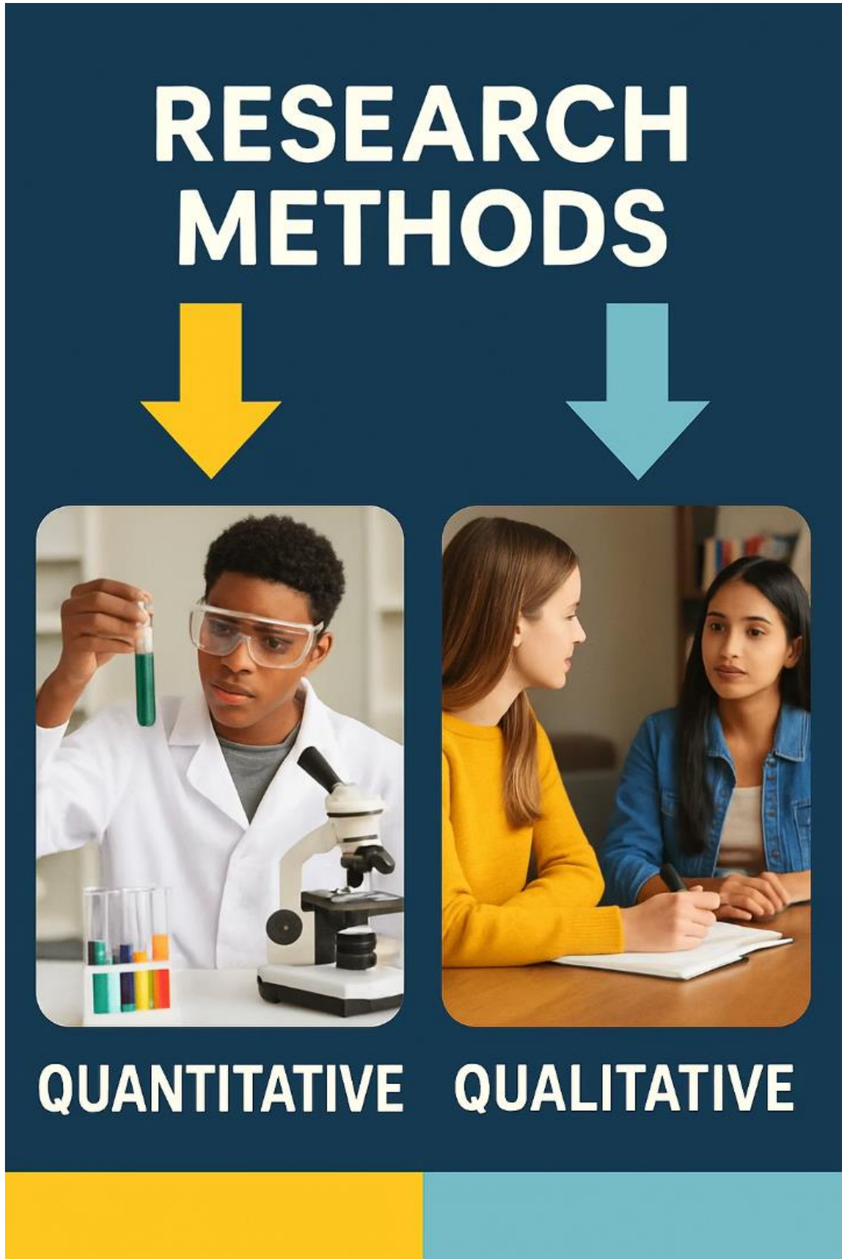
- Your data collection tools / techniques



Choosing a Research Method

The first step in designing your project is selecting the right method to answer your question.

Methodology includes…

- Your overall approach / strategy:

Quantitative Methods

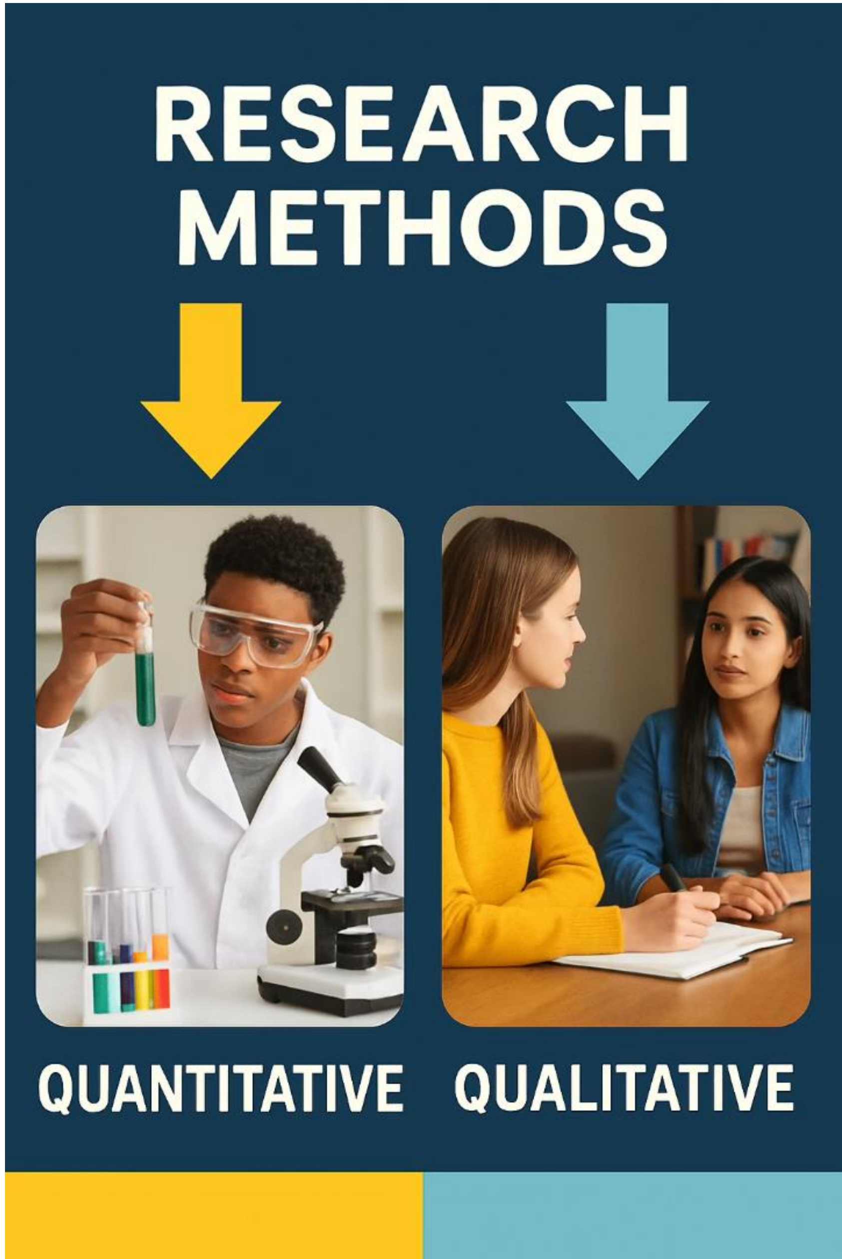
Focus on numbers and measurable data.

Good for questions about "how many," "how much," or testing relationships between variables.

Qualitative Methods

Focus on descriptions and non-numerical insights.

Good for exploring experiences, opinions, and understanding "why" or "how."



Choosing a Research Method

The first step in designing your project is selecting the right method to answer your question.

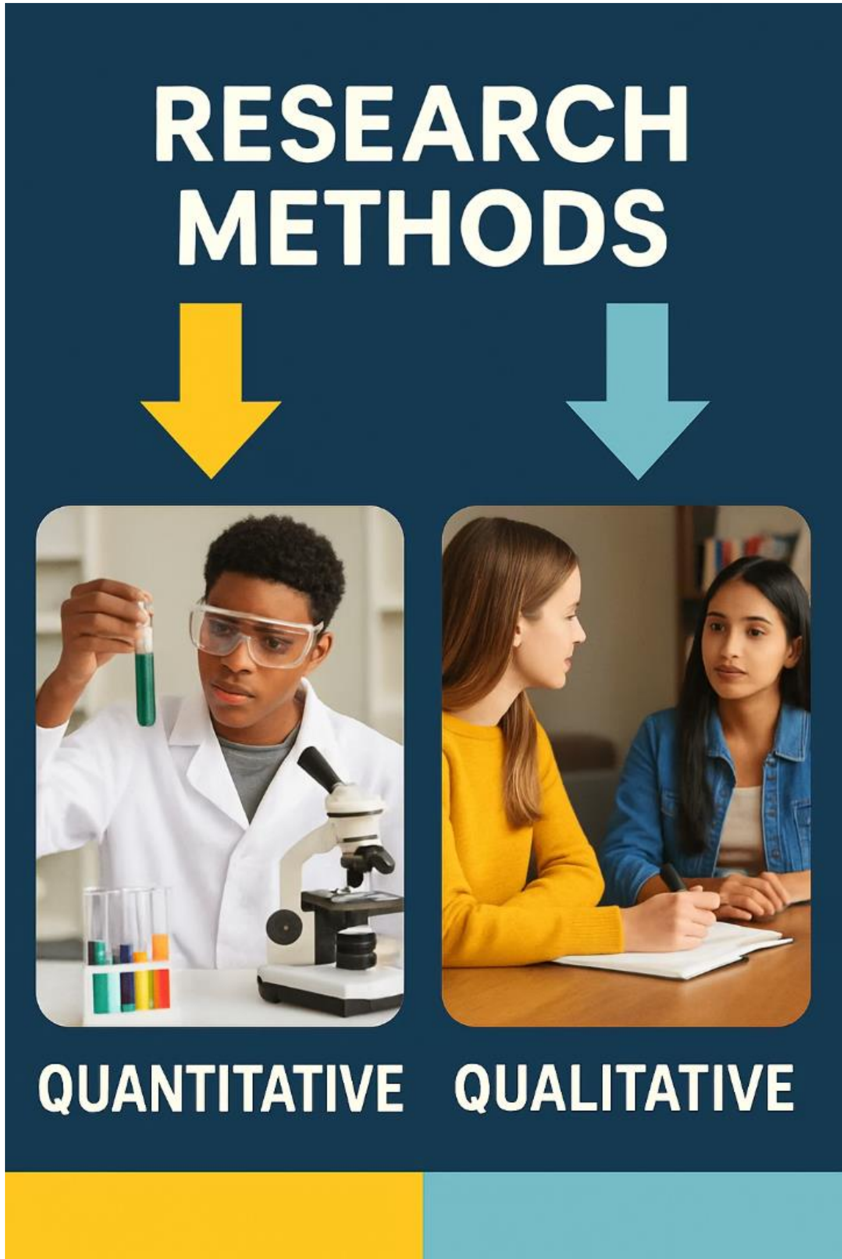
Methodology includes…

- Your overall approach / strategy:
  Qualitative. Quantitative. Mixed.

- Your data collection tools / techniques

HERE YOU HAVE A COUPLE DIFFERENT OPTIONS!

- You can use already collected data: Secondary Research
- You can collect your own data using one or multiple methods. Ex: You can conduct an experiment and/or make natural observations. (Next)

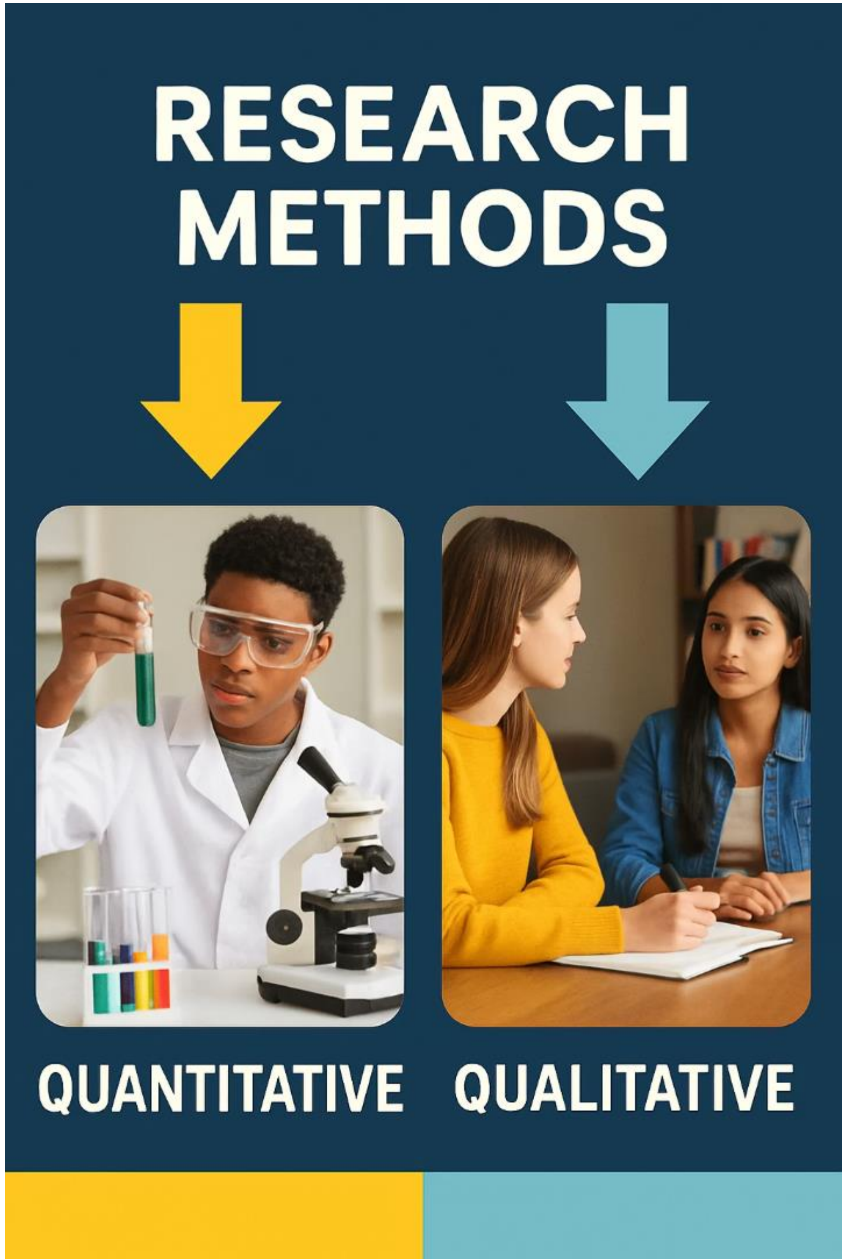


Choosing a Research Method

The first step in designing your project is selecting the right method to answer your question.

Methodology includes…

- Your overall approach / strategy:
  Qualitative vs. Quantitative
- Your data collection tools / techniques
  - Experiment
  - Surveys
  - Interview
  - Focus Groups
  - Observation
  - Case Study



Experiments: Testing Cause and Effect

When to Use

Best for questions about cause-and-effect relationships.

- You can manipulate variables
- You need to control other factors
- Your question starts with "What is the effect of...?"

Mostly quantitative

Key Components

- Independent variable (what you change)
- Dependent variable (what you measure)
- Control group (for comparison)
- Random assignment (reduces bias)

Example: Testing if sunlight affects plant growth by placing identical plants in different light conditions.

Quasi-Experiments: Real-World Testing

Definition

Similar to experiments but without random assignment or full control over all factors.

When to Use

When true experiments aren't feasible or ethical, but you still want to test relationships between variables.

Limitations

Harder to establish causation because other factors might influence results.

Mostly quantitative

Surveys: Gathering Opinions and Data

Broad Reach

Collect information from many people quickly.

Versatile Questions

Use multiple-choice, scales, or open-ended questions.

Quantifiable Results

Easily analyze and visualize responses.

Multiple Formats

Distribute on paper, online, or in person.

Common to use in both quantitative and qualitative studies

Interviews and Focus Groups: In-Depth Insights

Interviews

One-on-one conversations to explore personal experiences and perspectives.

- Detailed individual responses
- Follow-up questions possible
- Good for sensitive topics

Focus Groups

Guided discussions with several participants to gather multiple viewpoints.

- Group interaction generates ideas
- See how opinions form and change
- Efficient for hearing from many people

*Both methods provide rich **qualitative data** that numbers alone can't capture.*

Observation:
Watching Real Behavior

Tomorrow we continue with other methods of data collection!

Natural Settings

r as it naturally occurs without intervention.

nstructured

r specific behaviors or take open-ended notes

u see.

rent times to capture variation in behavior patterns.

Minimize Observer Effect

Be discreet so your presence doesn't change the behavior you're studying.

Can be utilized in quantitative or qualitative studies

Homework

- Handout 3 “Literature Review & Research Methodology” - Review what you learned about lit review, refine your research questions and hypothesis, start reflecting on your methodology.

- Diving into the literature and keeping track of your resources through “Source Tracker”. Cover 1-2 articles today, and share your notes with one another.

- Look into your data-set to make sure that your question and hypothesis can be studied with the information provided there.

EXTRA SLIDES FOR DAY 3

Activity: Peer Review

Form Pairs

Partner with another student to exchange work.

Review Questions

Is the question clear, focused, and researchable? Suggest improvements.

Review Hypotheses

Is the hypothesis testable? Does it clearly connect variables? Is reasoning sound?

Your Turn: Refine Based on Feedback

Consider Feedback

Review the suggestions from your partner.

Decide which feedback to incorporate.

Revise

Rewrite your research question if needed.

Refine your hypothesis to be clearer or more testable.

Make sure variables are clearly identified.

Activity: Literature Mapping!

Let’s see what other researchers have found about this issue?

- A package of articles
- Handout: Source Tracker

Quickly skim the available articles, list 2-3 articles that seem most relevant.